

CONFIDENTIAL — BASELINE REPORT



NEVARA

MONITORING, REPORTING & VERIFICATION PLATFORM

Restoration Intelligence Assessment Report

Satellite-Based Ecological Feasibility & Monitoring Assessment

PROJECT

Handenahalli Lake Restoration Baseline Assessment

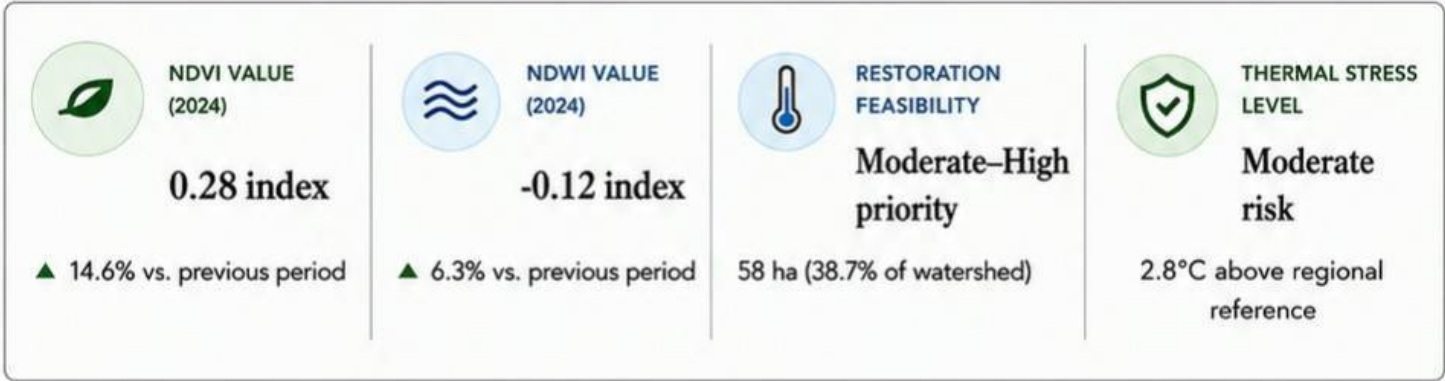
PREPARED FOR

SayTrees Environmental Trust

MAY 2026 • VERSION 1.0

Executive Summary

NEVARA’s satellite-based MRV intelligence framework was utilized to conduct a multi-dimensional ecological baseline assessment of Handenahalli Lake and its surrounding environmental influence zone. This executive summary synthesizes findings across vegetation health, hydrological dynamics, thermal characteristics, and restoration feasibility to inform strategic decision-making for the SayTrees Environmental Trust restoration programme.



The integrated baseline assessment indicates a moderately degraded ecosystem with high restoration potential. Key indicators show early signs of recovery following recent conservation interventions, creating a strong foundation for targeted restoration actions.

METHODOLOGY

Multi-spectral satellite analysis (Sentinel-2, Landsat 8/9), GIS-based spatial modelling, Temporal change detection (2019–2025), IPCC-aligned MRV framework.

CLASSIFICATION

Confidential — Baseline Report – NEVARA MRV Platform
Version 1.0 – May 2026

Site Overview & Location Intelligence

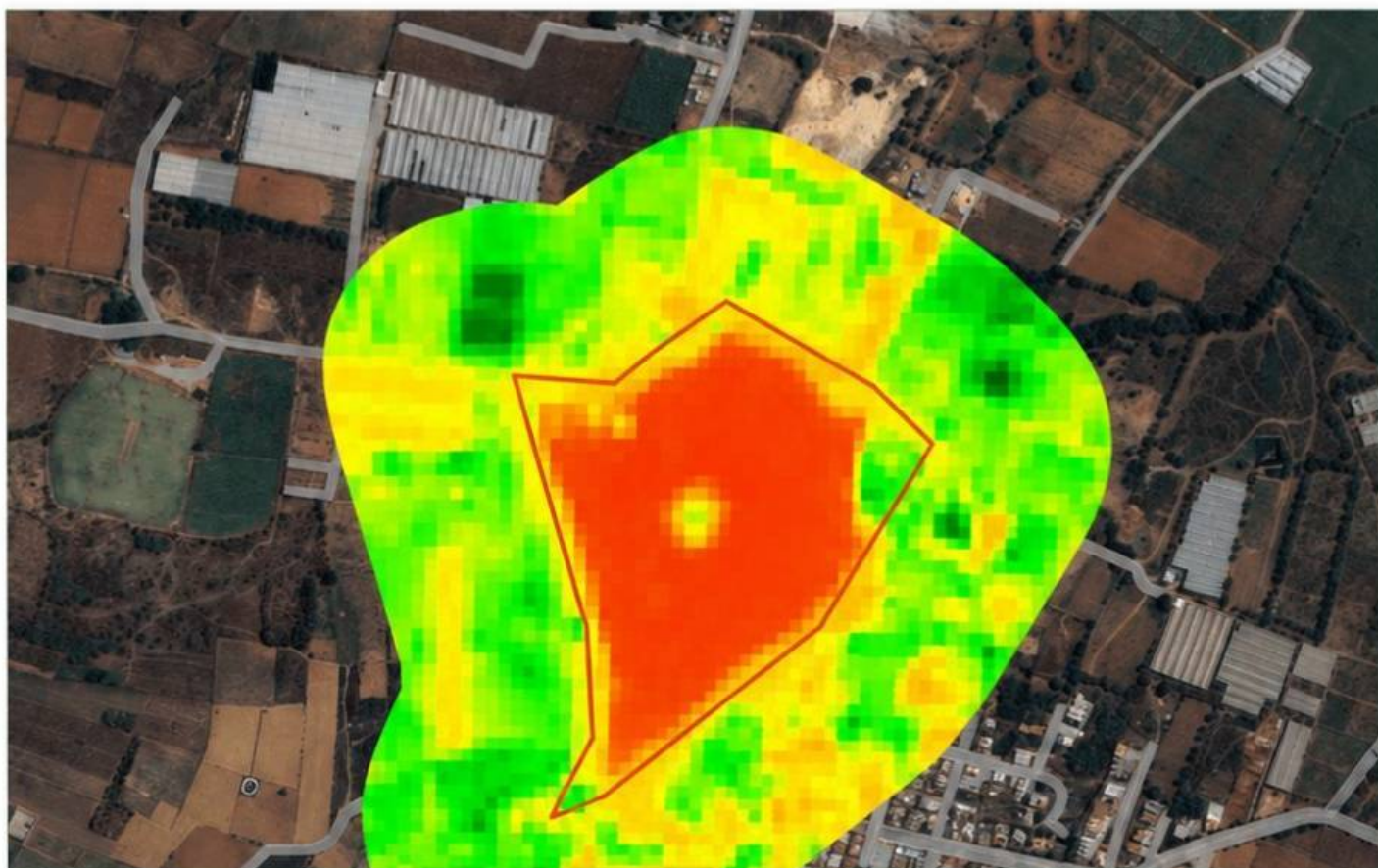


FIGURE 2.1

NDVI VEGETATION HEALTH ANALYSIS (2025)

Sparse to moderate vegetation density across the study area; productive canopy concentrated in the northeastern catchment.

Sentinel-2 MSI
10m resolution • May 2025

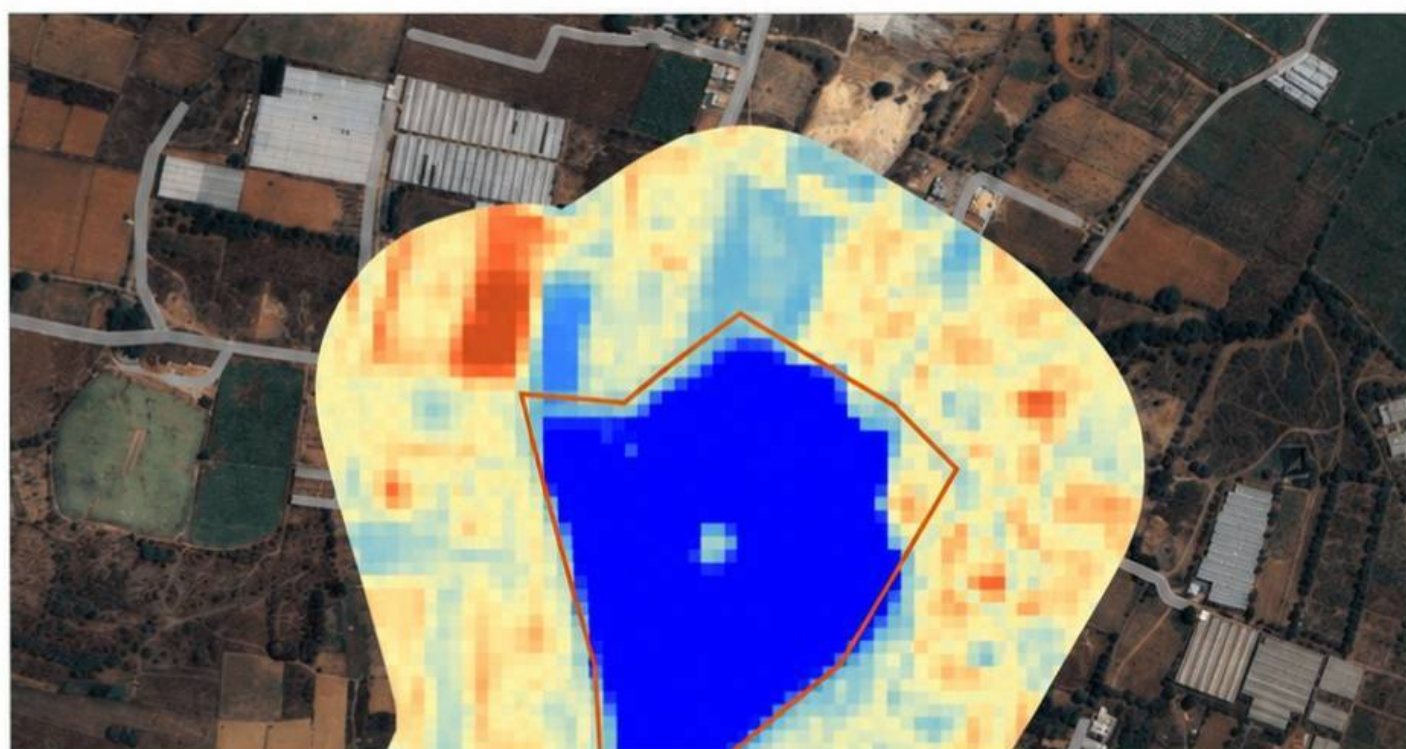


FIGURE 2.2

NDWI MOISTURE INDEX ANALYSIS (2025)

Surface moisture retention is low to moderate; core waterbody shows improved seasonal water persistence.

Sentinel-2 MSI
10m resolution • May 2025

Methodology & Satellite Analytics Framework

This assessment employs an integrated multi-sensor satellite remote sensing and GIS-based spatial analytical framework designed to deliver comprehensive ecological intelligence. The methodology follows a seven-stage workflow—from data acquisition through to MRV-aligned interpretation—ensuring scientific rigour and compliance with international environmental reporting standards.

01

Satellite Data Acquisition

Multi-sensor satellite imagery sourced from Copernicus Open Access Hub and USGS EarthExplorer. Primary sensors: Sentinel-2 MSI (Level-1C & Level-2A) and Landsat 8/9 OLI/TIRS (Collection 2, Level-2).

- Sentinel-2 MSI — 13 spectral bands, 10m–60m
- Landsat 8/9 OLI — 11 spectral bands, 15m–100m
- Temporal coverage: 2019–2025 (7-year archive)
- Cloud cover threshold: <5% scene selection criteria

02

Preprocessing & Atmospheric Correction

Radiometric calibration, atmospheric correction, and cloud masking applied to ensure consistent, analysis-ready data products across all temporal epochs.

- Sentinel-2: Sen2Cor processor (L2A generation)
- Landsat: LaSRC atmospheric correction algorithm
- Fmask algorithm for cloud, cloud shadow & snow masking
- Pan-sharpening: Gram-Schmidt method (15m resolution)

03

Spectral Index Computation

Multi-spectral vegetation, water, and built-up indices computed for each temporal epoch to derive biophysical parameters and ecological indicators.

- NDVI — Normalised Difference Vegetation Index
- NDWI — Normalised Difference Water Index (McFeeters)
- EVI — Enhanced Vegetation Index (atmospheric correction)
- NDBI — Normalised Difference Built-up Index
- MNDWI — Modified Normalised Difference Water Index

04

Terrain & Hydrological Modelling

Digital Elevation Model (DEM) analysis for terrain characterization and hydrological network extraction using GIS-based spatial algorithms.

- SRTM DEM (30m) — slope, aspect, hillshade derivation
- Watershed delineation — pour point & flow accumulation
- Drainage network extraction — Strahler stream ordering
- Hydrological connectivity — cost-distance analysis

05

Restoration Suitability Assessment

Multi-criteria decision analysis integrating topographic, hydrological, soil, and ecological parameters weighted by expert judgement and literature review.

- Input layers: slope, aspect, soil type, NDVI, NDWI, LST
- Weighted overlay — Analytical Hierarchy Process (AHP)
- Priority zoning — High / Medium / Low classification
- Species suitability — native species distribution modeling

06

Temporal Change Detection

Multi-epoch comparative analysis for quantifying landscape transformation, degradation trajectories, and recovery trends across the assessment period.

- Post-classification comparison — land cover transitions
- Image differencing — NDVI/NDWI/LST delta mapping
- Change Vector Analysis (CVA) — magnitude & direction
- Recovery trajectory — greening rate & trend analysis

Ecological Index Analysis

Multi-spectral satellite indices were computed from Sentinel-2 imagery to quantify vegetation health, biomass density, and surface moisture dynamics across the Handenahalli Lake intervention zone. The following analytical maps and derived metrics form the biophysical foundation of the ecological baseline assessment.

NDVI VEGETATION HEALTH ANALYSIS (2025)



FIGURE 4.1

Sparse to moderate vegetation density across the study area; healthy patches concentrated in the eastern riparian corridor.

Sentinel-2 MSI
10m resolution • May 2025

NDWI MOISTURE INDEX ANALYSIS (2025)

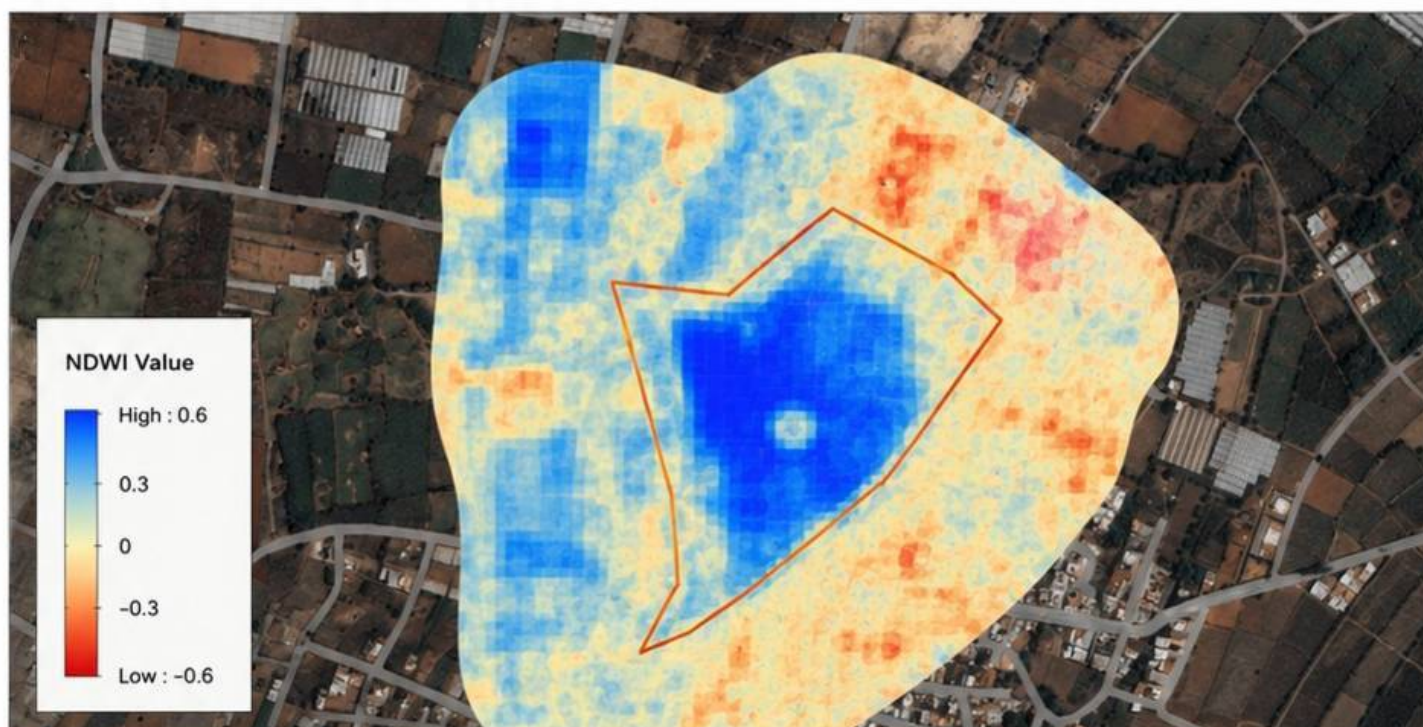


FIGURE 4.2

Surface moisture retention is low to moderate; core waterbody shows improved extent post-monsoon.

Sentinel-2 MSI
10m resolution • May 2025

Land Cover & Ecological Distribution

A detailed land cover assessment was conducted to characterise the spatial distribution of vegetation, surface water, and exposed soil across the Handenahalli Lake study area. This ecological classifying provides critical baseline data for understanding habitat configuration, degradation patterns, and restoration targeting.



FIGURE 5.1

Vegetation is concentrated in the eastern and northern riparian zones; western foreshore shows fragmentation.

Sentinel-2 MSI

10m resolution • May 2025

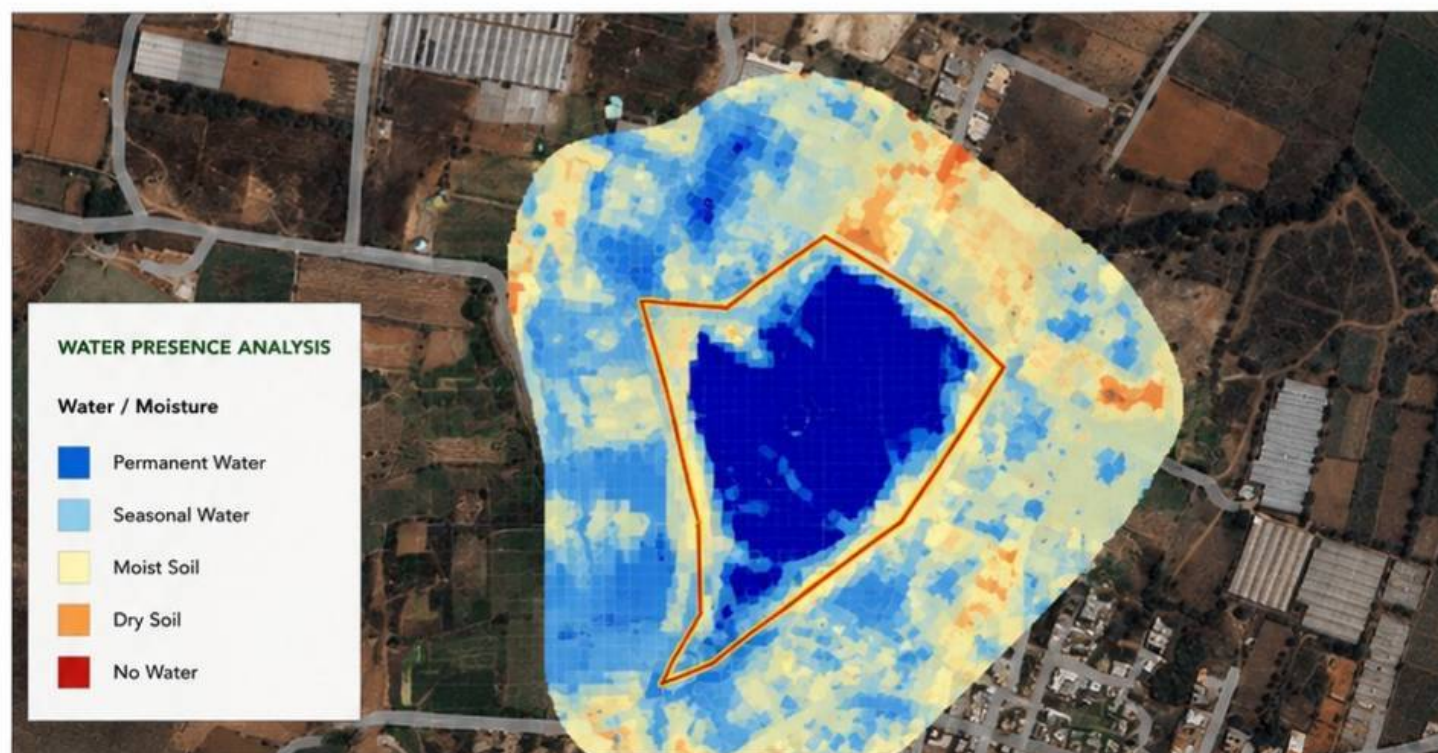


FIGURE 5.2

Core lake is the primary seasonal waterbody; peripheral areas experience moisture variability.

Sentinel-2 MSI

10m resolution • May 2025

Thermal & Climate Stress Analysis

Land Surface Temperature (LST) retrieval was performed using thermal infrared bands from Landsat 8/9 OLI/TIRS imagery to quantify surface thermal dynamics, identify stress zones, and assess the role of vegetation, water bodies, and microclimatic regulation across the Handenahalli Lake study area.



FIGURE 6.1

THERMAL HOTSPOT ANALYSIS

Thermal hotspots in western settlement fringe, exposed soil, and southern agricultural interface.

Landsat 8/9 TIRS
10m resolution • May 2025

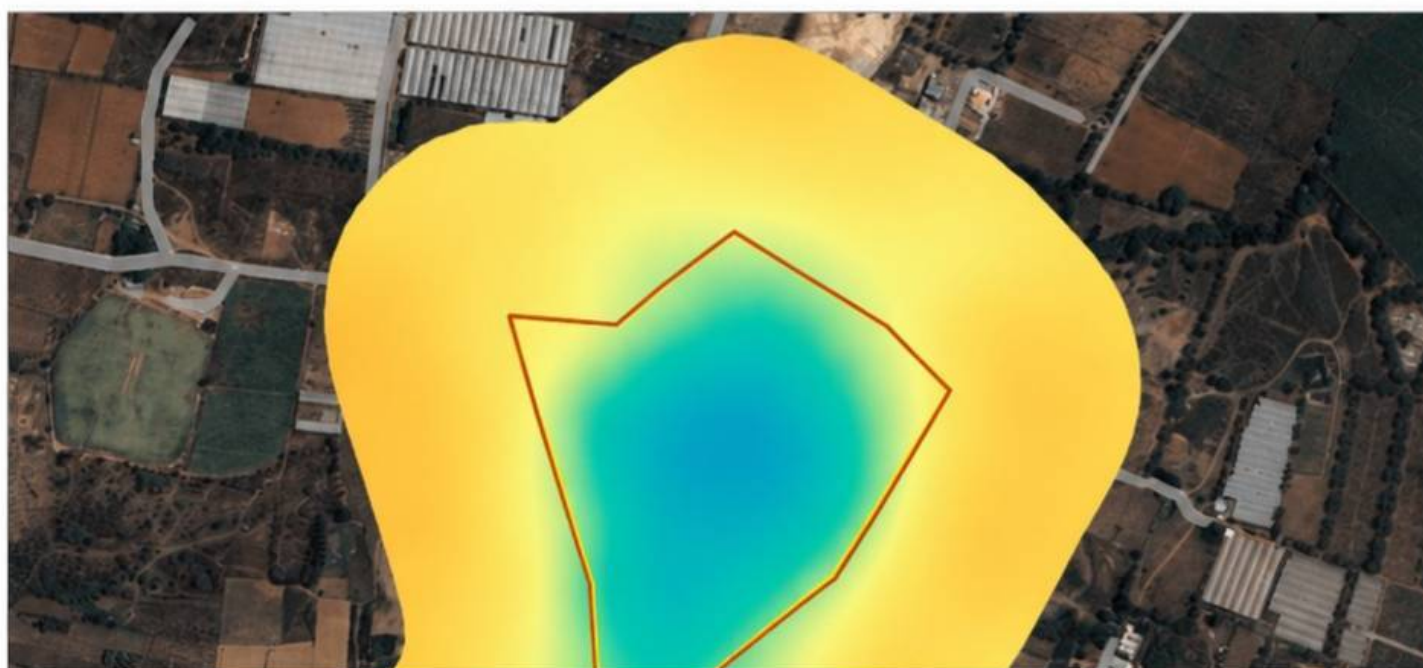


FIGURE 6.2

LAND SURFACE TEMPERATURE (LST)

Mean LST of 34.2°C — 2.8°C above regional reference, moderate microclimatic anomalies detected.

Landsat 8/9 TIRS
10m resolution • May 2025

Restoration Suitability Assessment

A multi-criteria restoration suitability analysis was conducted integrating topographic, hydrological, ecological, and land cover parameters to identify priority intervention zones and prescribe evidence-based rehabilitation strategies for the Handenahalli Lake ecosystem.

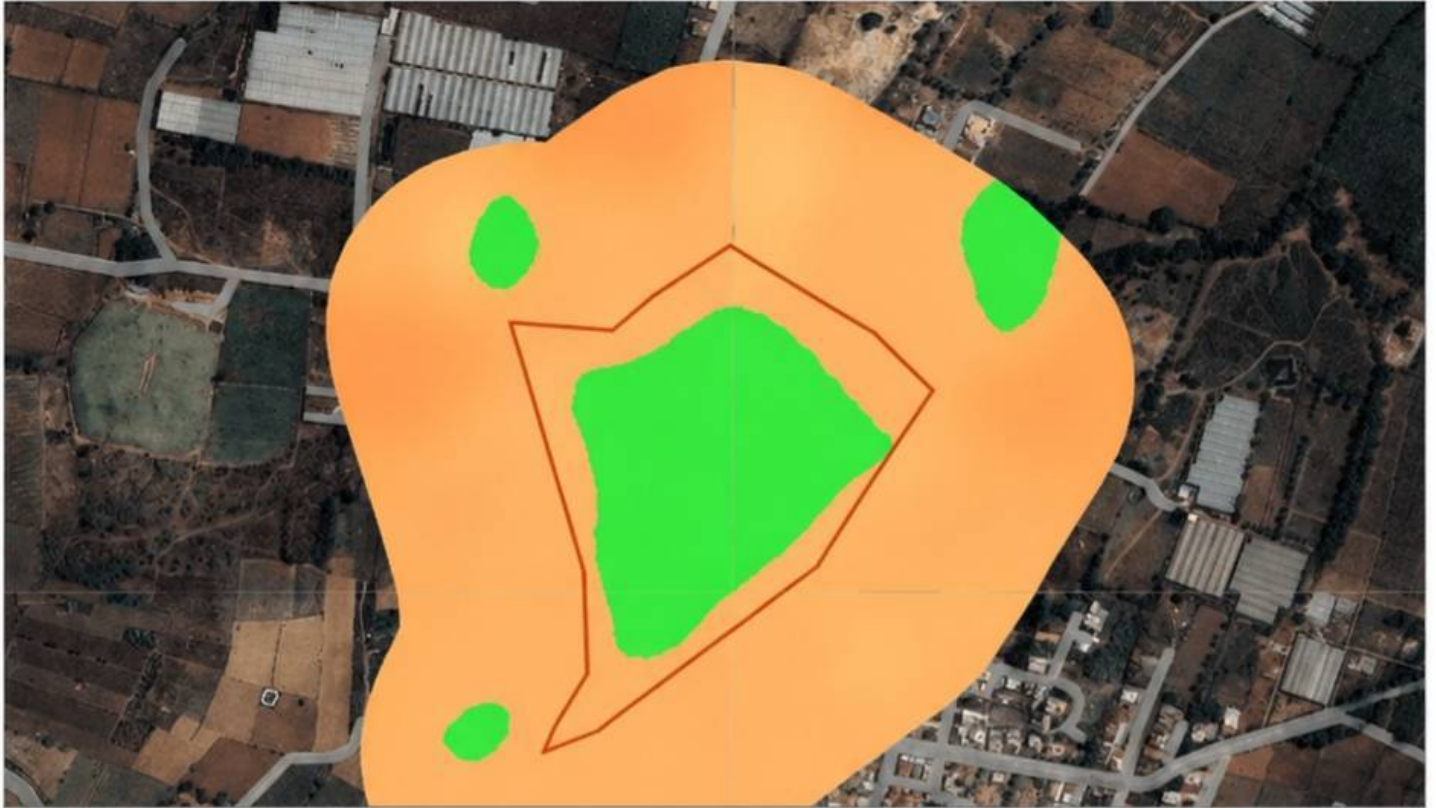


FIGURE 7.1

RESTORATION SUITABILITY ZONATION

AHP-weighted overlay for input criteria • 10m resolution

Sentinel-2 MSI

10m resolution • May 2025

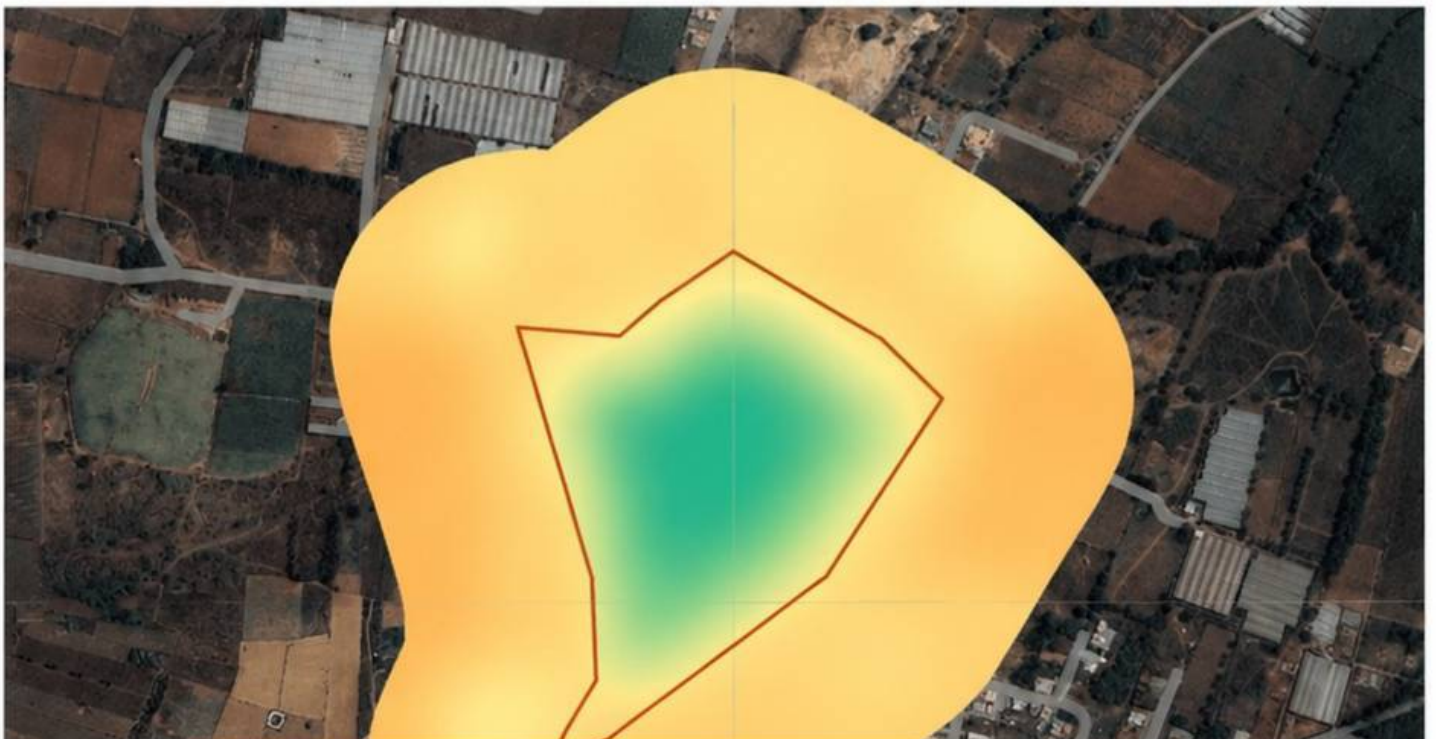


FIGURE 7.2

RESTORATION PRIORITY HEATMAP

Suitability intensity for restoration intervention planning

Sentinel-2 MSI

10m resolution • May 2025

Terrain & Hydrological Intelligence

Digital Elevation Model (DEM) analysis and hydrological network extraction were performed to characterize terrain morphology, slope stability, and surface drainage dynamics across the Handenahalli Lake watershed. This intelligence informs erosion risk assessment, and catchment-scale restoration planning.

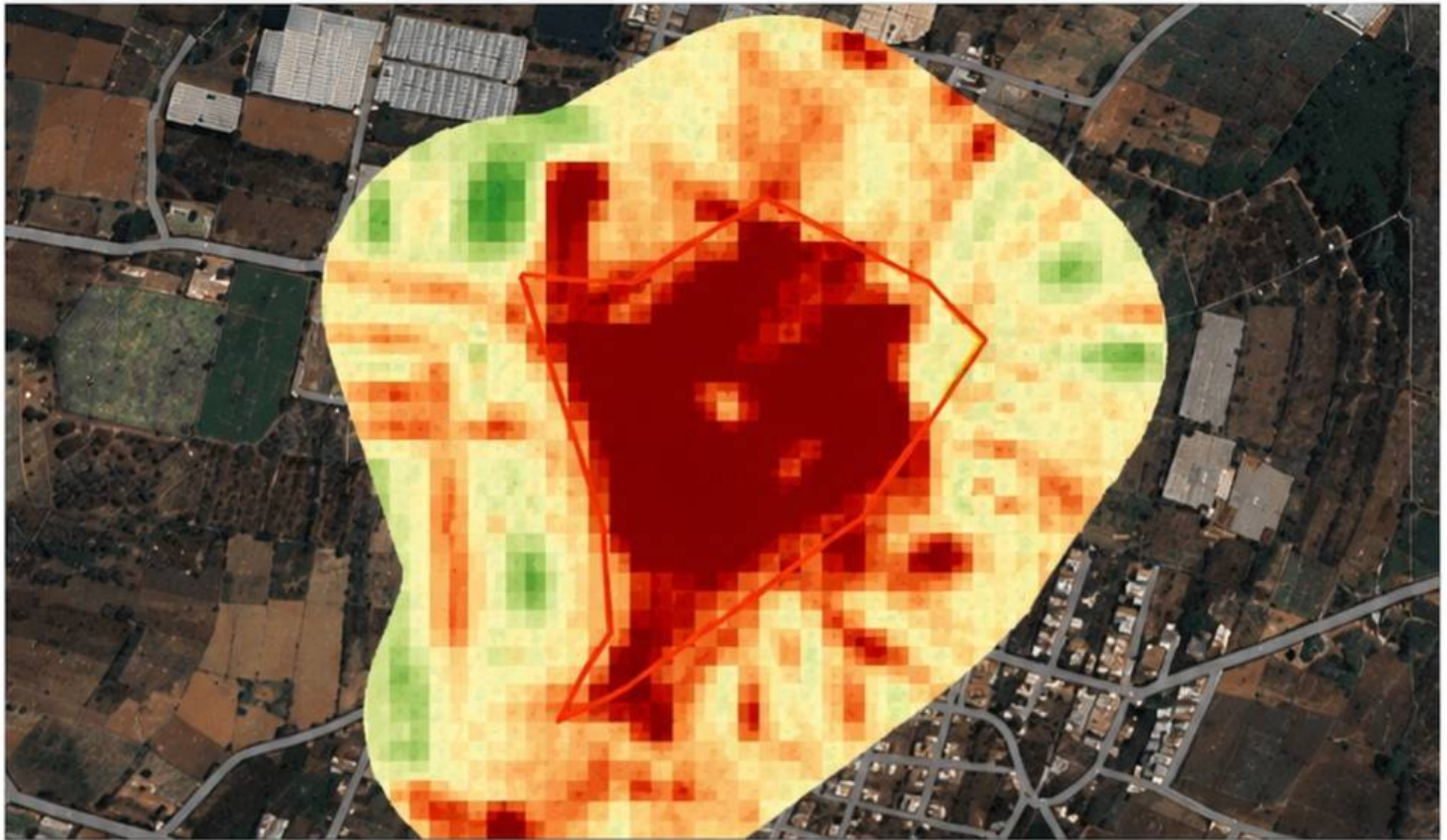


FIGURE 8.1

DIGITAL ELEVATION MODEL (DEM)

SRTM 30m • Elevation range: 820–940m AMSL

Sentinel-2 MSI
10m resolution • May 2025

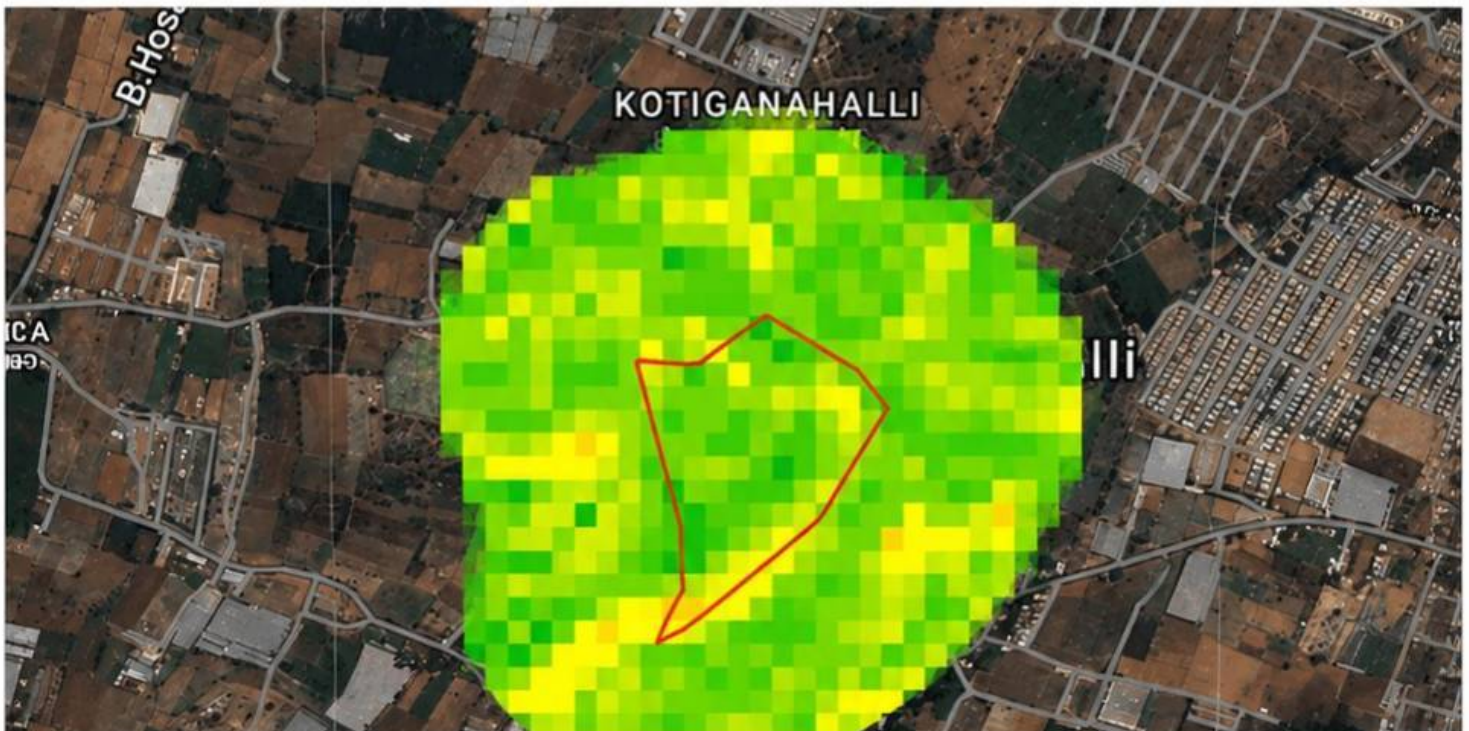


FIGURE 8.2

HYDROLOGICAL STRESS INDEX

Derived from DEM • Flow accumulation & slope analysis

Sentinel-2 MSI
10m resolution • May 2025

Temporal Change Detection

Multi-temporal analysis was conducted comparing spectral indices and land cover classifications between the 2020 baseline and 2025 assessment epochs to quantify landscape transformation, degradation trajectories, and recovery trends across the Handenahalli Lake study area.



FIGURE 9.1

NDVI TEMPORAL VEGETATION CHANGE (2020–2025)

Image differencing: Greening-browning trajectory · 10m resolution

Sentinel-2 MSI
10m resolution · May 2025

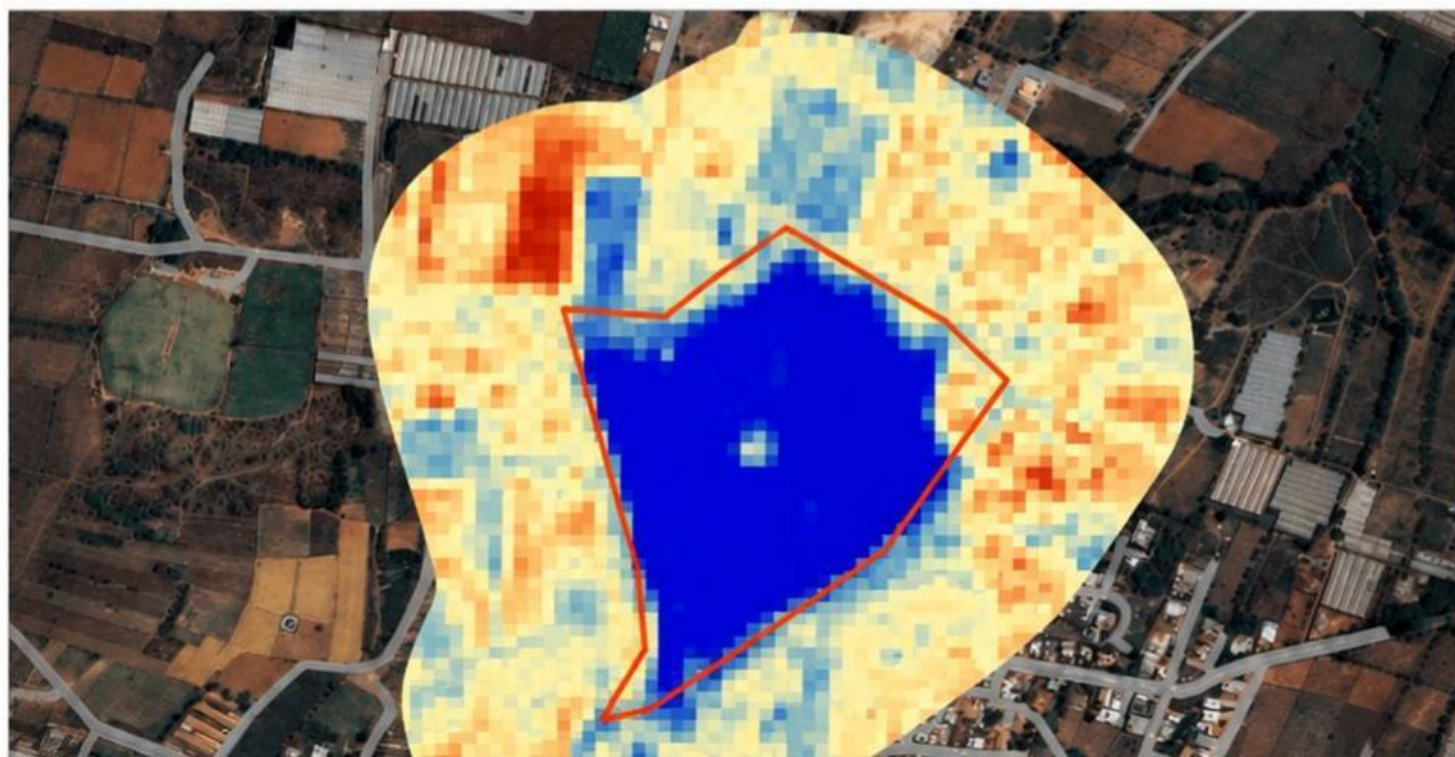


FIGURE 9.2

NDWI TEMPORAL WATER CHANGE (2020–2025)

Surface moisture variation and hydrological retention change analysis · 10m resolution

Sentinel-2 MSI
10m resolution · May 2025

Multi-year satellite observation reveals progressive ecological transformation across the Handenahalli Lake influence zone. Temporal imagery indicates gradual vegetation fragmentation, increasing peripheral exposure, and expanding anthropogenic pressure.

	● 2019 STABLE WATER PERSISTENCE BASELINE Healthy lake extent with intact riparian fringe and minimal anthropogenic disturbance.
	● 2020 INITIAL SHORELINE CONTRACTION SIGNALS First detectable reduction in waterbody perimeter with early vegetation thinning along the western margin.
	● 2021 PERIPHERAL VEGETATION FRAGMENTATION EMERGING NDVI decline accelerates in northeastern catchment; isolated vegetation patches become more distinct.
	● 2022 INCREASED EXPOSED SEDIMENT VISIBILITY Bare soil index elevates in southern agricultural interface; erosion signatures become apparent.
	● 2023 URBAN-EDGE ECOLOGICAL PRESSURE INTENSIFICATION Built-up expansion along southern corridor reaches critical proximity to lake buffer zone.
	● 2024 REDUCED HYDROLOGICAL BUFFERING PATTERNS Monsoon recovery improves moisture index but seasonal dry-down intensity increases.
	● 2025 MODERATE DEGRADATION WITH RESTORATION POTENTIAL Degradation plateau observed; intervention window remains viable for ecosystem recovery.

TRAJECTORY SUMMARY

The 7-year temporal sequence reveals a progression from stable ecological baseline (2019–2020) through degradation acceleration (2021–2023), followed by a potential plateau phase (2024–2025).

SATELLITE SENSOR

Sentinel-2 MSI · Landsat 8/9 OLI · Annual dry-season composites · Cloud-free mosaics · Consistent acquisition window (January–March)

MONITORING FRAMEWORK

Annual epoch comparison · Normalised spectral indices · Visual interpretation · Change matrix validation · NEVARA MRV temporal protocol

ASSESSMENT PERIOD

2019–2025 · 7 continuous years · 7 observation epochs · Seasonal consistency · Baseline: 2019 · Current: 2025

The integrated satellite-based ecological intelligence assessment indicates that the Handenahalli Lake landscape is currently experiencing moderate ecological degradation driven by vegetation fragmentation, hydrological stress, exposed sediment expansion, and increasing anthropogenic pressure. Despite these challenges, the environmental system retains significant restoration potential provided that phased ecological rehabilitation and long-term MRV monitoring frameworks are implemented within the current intervention window.

Priority: High

Complexity: Moderate

Recovery: Moderate-High

Sustainability: Strong

Phased Implementation Roadmap

01

Phase 1 — Immediate Stabilization

0–12 Months

- Check dams & contour trenches
- Silt traps at inflow points
- Shoreline reinforcement
- Runoff detention basins

02

Phase 2 — Ecological Recovery

1–3 Years

- Native riparian plantation
- Wetland vegetation recovery
- Biodiversity habitat structures

03

Phase 3 — Long-Term Monitoring

3–10 Years

- Quarterly satellite monitoring
- Annual biodiversity audits
- Carbon & hydrological tracking

Intervention Priority Matrix

INTERVENTION	ECOLOGICAL IMPACT	IMPLEMENTATION URGENCY	MONITORING REQUIREMENT
Erosion Control & Sediment Stabilization	High	Immediate	Quarterly
Runoff Diversion Structures	High	Immediate	Seasonal
Shoreline Reinforcement	High	12 Months	Annual
Native Species Plantation	High	1–2 Years	Annual
Riparian Buffer Restoration	High	1–2 Years	Annual
Wetland Vegetation Recovery	Medium	1–3 Years	Annual
Biodiversity Enhancement	Medium	2–3 Years	Annual

EXECUTIVE SYNTHESIS

Multi-temporal ecological assessment indicates moderate but reversible landscape degradation within the Handenahalli Lake watershed. Satellite-derived NDVI, NDWI, and LST trends confirm that targeted restoration intervention can significantly improve vegetation recovery, hydrological resilience, and thermal regulation.

RESTORATION OPPORTUNITY SUMMARY

Approximately 58 hectares (38.7%) of the 150-hectare watershed has been classified as high-priority restoration terrain. Riparian rehabilitation, shoreline stabilization, and wetland recovery interventions present strong ecological and long-term monitoring feasibility.

LONG-TERM ECOLOGICAL OUTLOOK

Projected ecological recovery trajectories indicate measurable improvement in vegetation density, sediment control, and hydrological stability over a 10-year implementation horizon. Continuous MRV monitoring will support adaptive restoration management and impact verification.

SUSTAINABILITY POTENTIAL & MRV SIGNIFICANCE

The integrated NEVARA MRV framework enables standardized ecological monitoring using satellite intelligence, GIS analytics, and field verification workflows. The framework supports transparent reporting aligned with Verra VCS-aligned freshwater restoration and adaptive governance protocols.

This assessment affirms that the Handenahalli Lake ecosystem, while facing significant ecological pressure, retains the fundamental capacity for recovery. The convergence of satellite intelligence, GIS analytics, and MRV compliance frameworks provides an unprecedented foundation for evidence-based restoration.

Operational Decision-Support Intelligence

This operational intelligence layer translates satellite-derived ecological analytics into actionable field-level restoration decisions. By integrating intervention sequencing logic, catchment analysis, and shoreline condition classification, this framework provides restoration practitioners with prioritised, terrain-informed implementation guidance with ecological urgency and operational feasibility.

Intervention Effort–Impact Matrix

HIGH IMPACT - LOW EFFORT		Phase Priority
Native shoreline buffer planting Phase 1 Impact: High Effort: Low Rapid ecological stabilization with immediate visible results within one monsoon cycle		
Sediment control barriers Phase 1 Impact: High Effort: Low Cost-effective intervention for reducing lake sedimentation load by 40-60%		
HIGH IMPACT - MODERATE EFFORT		Phase 1–2
Runoff interception bunding Phase 1 Impact: High Effort: Moderate Critical for sediment control and hydrological regulation in high-drainage zones		
Wetland fringe strengthening Phase 2 Impact: High Effort: Moderate Enhances aquatic biodiversity and hydrological buffering capacity		
HIGH IMPACT - HIGH EFFORT		Phase 2–3
Peripheral canopy restoration Phase 2 Impact: High Effort: High Long-term ecological recovery requiring site preparation and sustained maintenance		
MODERATE IMPACT INTERVENTIONS		Support Actions
Invasive vegetation clearance Phase 1 Impact: Moderate Effort: Moderate Essential precursor to native planting for reducing competition pressure		
Thermal buffer green corridors Phase 3 Impact: Moderate Effort: High Long-term microclimate regulation requiring corridor connectivity planning		
Community riparian planting Phase 2 Impact: Moderate Effort: Low Community engagement with high social co-benefit and moderate ecological impact		

Catchment Contribution Zones



- High-Contribution Runoff Corridors**
- **Operational Pressure:** High operational pressure — concentrated flow from western catchment
 - **Ecological Interpretation:** Drainage density exceeds 2.5 km/km with direct sediment delivery to lake waterbody
 - **Intervention Implication:** Immediate intervention required to reduce erosion and sediment transport

Monitoring Triggers & Adaptive Restoration Governance

Adaptive restoration governance integrates continuous satellite-derived ecological observation with predefined intervention thresholds and escalation protocols. This framework enables restoration teams to detect ecological degradation signals before they become irreversible — transforming monitoring data into actionable restoration governance through automated trigger detection, risk classification, and phased intervention deployment.

Ecological Trigger Threshold Dashboard

 NDVI WARNING 0.18 Healthy Warning Critical > 0.30 0.15 – 0.30 < 0.15 Monitoring Below 0.15 triggers urgent canopy intervention & replanting deployment Quarterly
 NDWI WARNING -0.12 Healthy Warning Critical > -0.05 -0.20 to -0.05 < -0.20 Monitoring Below -0.20 triggers hydrological assessment & wetland recharge intervention Quarterly
 LST WARNING 32–5 °C Healthy Warning Critical < 30°C 30–35°C > 35°C Monitoring Above 35°C triggers thermal buffer corridor deployment & microclimate intervention Monthly (Summer)
 Shoreline Stability WARNING 62% stable Healthy Warning Critical > 75% 50–75% < 50% Monitoring Below 50% triggers shoreline reinforcement, riprap deployment & buffer planting Quarterly

This report has been developed using satellite-derived ecological intelligence methodologies, GIS-based analytical workflow, and environmental monitoring protocols aligned with internationally recognized restoration and MRV frameworks.

Compliance Framework Alignment

STANDARD / FRAMEWORK	ALIGNMENT STATUS	SCOPE & APPLICATION
Vera VCS VM0042 / VM0047	COMPATIBLE	Improved Land Management and wetland investing methodologies
IPCC Tier 2/3	COMPATIBLE	National Greenhouse Gas Inventory guidelines for land use sectors
ISO 14064-2	ALIGNED	Greenhouse gas accounting and verification at project level
GHG Protocol	ALIGNED	Corporate accounting and reporting standard for emissions
CCB Standards	REFERENCED	Climate, Community & Biodiversity co-benefit certification
NEVARA MRV Framework	FULLY INTEGRATED	Proprietary satellite-based monitoring, reporting & verification platform

Satellite Data Provenance & Traceability

DATA SOURCE	RESOLUTION	PROVIDER	ANALYTICAL USAGE
Sentinel-2 MSI	10m (VHA) / 20m (SWIR)	Copernicus ESA	Primary vegetation & water indices
Landsat 8/9 OLI/TIRS	30m (multispectral) / 100m (thermal)	NASA / USGS	LST retrieval & temporal baseline
SRTM DEM	30m	NASA / USGS	Terrain & hydrological modeling
Survey of India Toposheets	1:50,000	SOI	Administrative boundary validation

Analytical Reproducibility & Data Provenance Declaration

All analytical outputs presented in this report are fully reproducible through documented workflows. Raw satellite imagery is archived with unique scene identifiers and accessible via Copernicus Open Access Hub and USGS EarthExplorer. GIS project files, processing scripts, and intermediate data products are maintained within the NEVARA MRV platform with full version control and audit trail. Ground truth validation data is georeferenced and stored in the project geodatabase. All spectral indices, classification outputs, and statistical analyses include metadata documenting algorithm parameters, confidence intervals, and accuracy assessments.